CrystalRL — Crystal Clear Reinforcement Learning

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Abstract

Post-hoc explanation of a trading policy answers the wrong question: it tells a story *about* a black box while the box goes on being one. We report the construction of an agent that is legible *by* construction — and, critically, the measurement discipline that keeps "legible" from being a vibe. The paper traces one concrete journey: from **CHRL**, a constrained hierarchical PPO trader whose three decision layers are readable but whose core is a 64-dimensional unnamed latent with ∼214 tuning knobs, to **CRYSTAL-1**, whose memory is a *named* probability vector over market regimes (a K-simplex the operator can write to), whose policy head *is* an ≤8-leaf soft decision tree at full return parity (gap −0.018, paired $p{\approx}0.84$), and whose entire command surface is five named levers under a certified write-governance stack. On the identical scalar — CrystalScore = Faithfulness × Simulatability × Stability at parsimony budget $K{\leq}9$ — the journey moves the agent from 0.15 to 0.94, and the entire gap is Simulatability: a short named story reproduces CRYSTAL-1's behavior and cannot compress CHRL's latent. We catalogue the four families of behavioral influence the design admits (reward shaping, teachers, belief interventions, architecture), each with a measured outcome and one honest refusal. We state the nulls that bound the claims: the tension between return and legibility is *certified* on hard substrates; CRYSTAL-1's command surface is polygon-proven while its predecessor still leads as a deployed system; and composition of commands is non-additive with sign flips, which is why governance is machinery, not policy. The paper closes with a research program of twelve falsifiable hypotheses for the interpretability track.

1 Introduction: the question behind "explainable"

When a reinforcement-learning agent trades real money, "why did it do that?" is not a curiosity — it is a regulatory demand, a debugging requirement, and (in our companion work on self-improving systems) a safety precondition: *an agent that changes itself may only be allowed to do so to the degree it can be read.* The dominant practice answers with post-hoc explanation: train a black box, then probe it — saliency, surrogate models, cluster the latent. Rudin [2019] names the flaw: for high-stakes decisions, do not explain the black box; build the model interpretable in the first place. The critique is well known. What is scarcer is a *worked example with a measurement discipline*: an agent built legible by construction, on a task where the black-box baseline is strong, with the legibility difference expressed as a number that both architectures can be scored on — and with the costs stated.

This paper is that worked example. It has the shape of a journey because we walked it in order: we first built the transparent-*layers* agent (CHRL), measured where its transparency stops, and then rebuilt the core so that the stopping point disappears (CRYSTAL-1). The destination is summarized in one scalar (Figure 2), but the scalar is the least of it; the contribution is the discipline that makes each factor of that scalar a measured, attackable claim.

Contributions:

1. **A final CHRL→CRYSTAL-1 comparison** (§2, §3, Table 1) — architecture, control surface, and behavior, on identical metrics; the predecessor is treated honestly (it still leads as a *deployed* system).

2. **CrystalScore and its measurement protocol** (§4): faithfulness as *proved causal* command-following (against an exogenous optimum, not a correlation), simulatability as a parity-constrained short story, stability as multi-seed replication — plus the MDL deficit as the non-saturating companion.

3. **A taxonomy of behavioral influence** (§5): reward shaping, teachers (behavior cloning / warm starts), belief interventions, and architecture — each exercised on CRYSTAL-1 with a measured outcome, including the intervention the safety bar *refused*.

4. **Governed composition** (§6): commands interact non-additively with sign flips (216-cell factorial); we describe the ledger, co-activation cap, manifold governor and writ ladder that turn this from a hazard into an operating envelope, with the first estimate of the machine's own false-accept rate (α_{machine}).

5. **The honest nulls** (§7): the certified return↔legibility tension; the polygon→market gap and the VoI gate that fences it; what post-hoc interpretability *did* deliver on CHRL (it was not nothing).

6. **A think-big research program** (§9): twelve falsifiable hypotheses, each with a kill test, for the interpretability track.

2 The predecessor: CHRL, or how far "transparent layers" go

CHRL (Constrained Hierarchical RL) was our best answer to interpretability *within* the standard recipe. Decisions are factored into three readable layers — a risk layer choosing cash-vs-risky exposure on a ∼20-day rhythm, a group layer routing capital through stock clusters, and a stock layer picking names weekly — wrapped in explicit guardrails: exposure pacing, confidence-scaled trade sizing, event triggers, top-K routing, and a buy gate requiring recovery evidence. Hierarchy as a legibility device has a long RL lineage [Sutton et al., 1999, Bacon et al., 2017]; the PPO core is standard [Schulman et al., 2017].

It works as a trader: in four-fold walk-forward validation the champion variant delivered mean validation return 10.05%, Sharpe 1.09, max drawdown −7.34%, mean cash weight 41.24%, and 0.84%/day L1 turnover. And its *post-hoc* interpretability program delivered real results: self-supervised primitive discovery over the frozen policy found behavioral motifs whose targeted edits improved a frozen 2022–23 rollout without retraining — promoting a context-specific primitive added +0.54% return / +0.049 Sharpe on the frozen test, with control-adjusted checks. Post-hoc analysis is not nothing; we state this against our own thesis.

But every question we most cared about hit the same wall, and the wall has coordinates. The core of CHRL — and of R6c, the line's frozen-tested champion — is a 64-dimensional *unnamed* latent. Memory is smeared through the network, so no bound exists on what a state change can touch. The control surface is ∼214

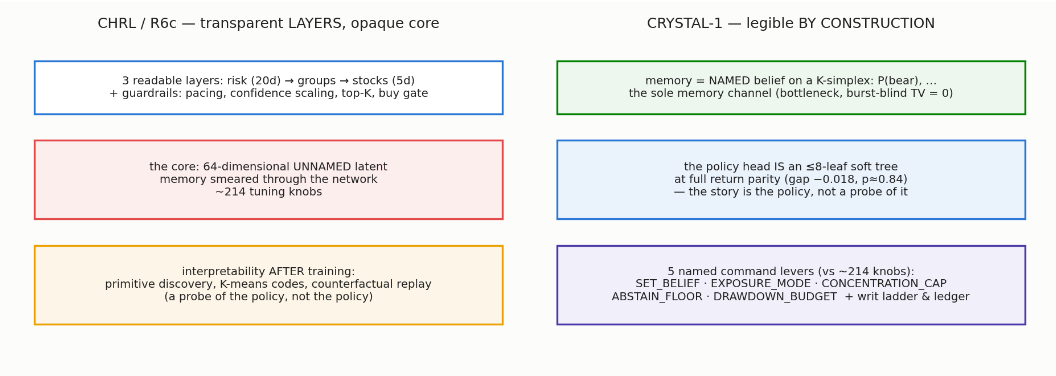

Figure 1: The journey in one picture. CHRL/R6c factors decisions into readable layers but keeps an unnamed 64-d latent core with ∼214 knobs, explained only after training. CRYSTAL-1 makes the explanation the mechanism: a named belief simplex as the sole memory channel, a soft tree as the policy itself, and five named command levers under certified governance.

engineering knobs, none of which is a semantic command. Steering means forcing the latent toward a post-hoc K-means centroid and checking that behavior moves *consistently* — a correlation, not a proof. There is no likelihood, so a command cannot be checked against the world (no way to ask "is this belief a *lie*?"). And when we later scored both architectures on the same simulatability metric, a $K \leq 9$ story reproduced only 0.244 of R6c's stance behavior — the latent is genuinely incompressible at that budget (§4). Transparent layers, opaque core (Figure 1).

3 CRYSTAL-1: legible by construction

CRYSTAL-1 rebuilds the core around three commitments.

1. Memory is a named probability vector. The agent's only recurrent state is a belief b_t on a K-simplex over market regimes — "P(bear)" is an actual posterior, in the POMDP sense of a belief state [Kaelbling et al., 1998], over Hamilton-style regimes [Hamilton, 1989]. *Named* is load-bearing: because coordinates have meanings, a write to the belief is a sentence ("believe bear at 0.8"), and because the belief is a bottleneck — the sole memory channel, with the raw burst feature structurally excluded from the head (total variation of the action distribution to the excluded feature = 0.0, 0/225 flips) — the blast radius of any memory intervention is bounded by construction.

2. The policy head is the story. The actor is a soft decision tree [Frosst and Hinton, 2017] over [belief, inventory, time] with at most 8 leaves, trained jointly (not distilled as an afterthought). The honest history: trained cold, the tree cost return (gap −2.57); with annealing −0.27; with a behavior-cloning warm start the gap is −0.018 — **full parity** with the MLP head (paired $p \approx 0.84$). This is the difference between a *probe* (post-hoc tree that approximates the policy) and a *constitution* (the tree IS the policy): there is no residual black box for behavior to hide in.

3. Commands are few, named, and governed. The ∼214-knob surface collapses to **five agent-facing levers**:

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SET_BELIEF · EXPOSURE_MODE · GROUP_CONCENTRATION_CAP
ABSTAIN_FLOOR · DRAWDOWN_BUDGET
```

(the concentration cap is *hard*, fixing the predecessor's soft-cap failure). Everything else is engineering-fenced. Commands compile to *certified writes* through a governance stack described in §6.

Two further properties come free from the design and are impossible for the predecessor. Because CRYSTAL-1 carries a world model with a likelihood, a commanded belief can be *checked against evidence*: a sharp-but-wrong belief shows excess predictive NLL 0.43 vs ≈0 for uncertain-but-right, detection 0.71 at 5% false alarms — and the naïve alternative, an entropy gate, points *backwards* (it would trust the confident lie). Authority tracks evidence, not confidence. And because writes are barycentric on the simplex, they are valid by construction: the refusal rate of the write primitive is 0, where the predecessor needed an OOD gate that shrinks or refuses off-manifold steps.

4 CrystalScore: measuring "legible" so it can lose

A legibility claim that cannot lose is marketing. We use

$$\textsc{CrystalScore} \;=\; \text{Faithfulness} \times \text{Simulatability} \times \text{Stability},$$

at a fixed parsimony budget $K \leq 9$, with each axis operationalized so that it *could have* come out low [Doshi-Velez and Kim, 2017, Jacovi and Goldberg, 2020, Lipton, 2018]:

- **Faithfulness** = does a belief write *causally* control action? Not a correlation: agreement with an *exogenous* belief-MDP optimum after the write — 0.80 for the corner PPO head, 0.835 for the soft-tree head; the read-side certificate additionally bounds the leak around the bottleneck (on-manifold TV 0.067, i.e. 5.3%; belief governs 83% of contexts). Interventional, in the spirit of Pearl [2009].
- **Simulatability** = can a $K{\leq}9$ named story reproduce the policy at *return parity*? (A story that reproduces a lobotomized policy is cheating; parity is part of the metric.)
- **Stability** = does the mechanism replicate across training seeds? (3/3 mechanism replication; 10-seed behavioral stability 1.0 on the current substrate.)

The companion metric is the **MDL deficit** [Rissanen, 1978]: the accuracy lost by forcing the explanation to be short. Unlike the product score it does not saturate; CRYSTAL-1's current MDL deficit is 0.0 — a short story loses nothing — which is precisely why hardening this number across substrates is the first task of the research program (§9).

Table 1 is the final comparison the journey was walked for.

5 The behavioral-influence toolbox: four levers, each measured

A legible agent is only useful if you can *change* it on purpose. We exercised four distinct families of influence on CRYSTAL-1's PPO heads — reward, teachers, interventions, architecture — and report one measured outcome per lever (Figure 3).

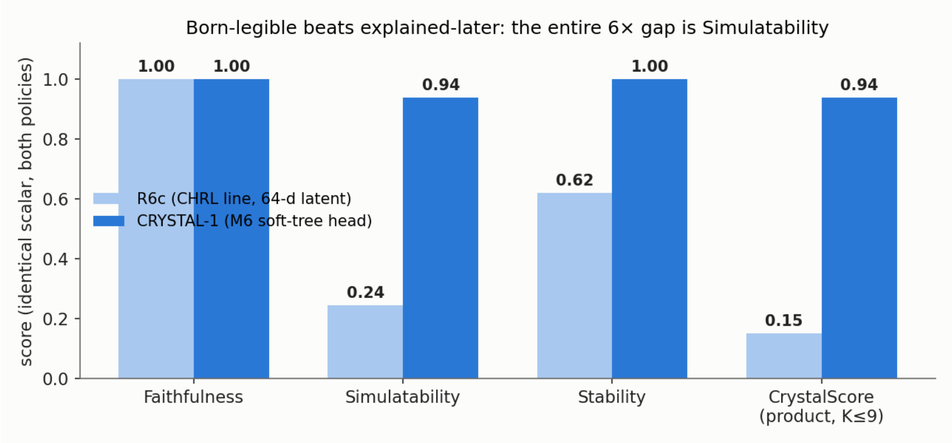

Figure 2: The identical scalar on both architectures. Faithfulness is 1.0 for both — *steering works on a black box too*; that axis alone cannot distinguish the designs. The 6× gap (0.151 → 0.938) is entirely Simulatability: a $K \leq 9$ named story reproduces CRYSTAL-1's policy and cannot compress R6c's 64-d latent stance (0.244). Alpha does not enter the score; the two axes are orthogonal by design.

R — reward shaping. The risk-targeted reward moves the drawdown dial reliably [Ng et al., 1999]: heads trained at different budgets deliver their budgets on held-out data. The honesty rider: the cold heads converge to near-*constant* exposure dials (0.25/0.5/0.75) — a drawdown-matched twin shows $z = 0.00$ timing skill. Reward shaping bought *risk placement*, not timing; on this substrate a dial is the optimum, and cold PPO re-discovered that from scratch (60k steps found no $\sim 2\sigma$ timing).

T — teachers. The cautionary tale first: the cold heads' argmax behavior looked sensible while the underlying preferences were nearly flat — the network was effectively untrained, and deterministic evaluation masked it. (For a behavioral scientist: inferring function from the argmax of a flat tuning curve.) The cure is a teacher: behavior cloning from the certified readable rule (87% agreement), used as a warm start, after which PPO fine-tuning *keeps* the teacher's behavior (warm = BC). The same mechanism closed the soft-tree parity gap ($-2.57 \to -0.018$, §3) — imitation as initialization, optimization as refinement [Ross et al., 2011, Rusu et al., 2016].

I — interventions. Writes to the named belief are the *do*-operator made concrete [Pearl, 2009]: fidelity 1.0 (the commanded belief is the realized belief), a sharp monotone dose-response (threshold ≈ 0.26 on the polygon), and ± 2-nat promote/suppress nudges that slide the agent between named behavior codes. The result we are proudest of is a refusal: a $+1$-nat promotion that would have degraded the certified objective was *blocked by the non-inferiority bar*. An intervention surface that cannot say no is a jailbreak surface.

A — architecture. Tree depth is a legibility dial: depth-2 yields a 4-leaf policy that remains legible end-to-end. Depth trades story size against expressiveness *ahead* of training — the one lever that changes what the agent *can* become rather than what it does.

	CHRL / R6c	CRYSTAL-1
Memory	64-d unnamed latent, smeared; blast radius unboundable	named K-simplex belief; sole channel; burst-blind (TV = 0)
Policy legibility	post-hoc codes over a continuous latent	the head *is* an 8-leaf tree at parity (-0.018, $p{\approx}0.84$)
Steering primitive	force latent toward a K-means centroid	named barycentric belief write
Causality of a command	correlation demo	*proved* vs exogenous optimum (0.80/0.835)
Command validation	none (no likelihood)	lie-detector: authority tracks NLL, not sharpness
Refusals / OOD	OOD gate; can refuse to zero authority	0 refusals by construction + kNN manifold governor
Control surface	$\sim$214 knobs	5 named levers + writ ladder + ledger
Composition governance	none live	measured sign-epistasis → calibrated ledger, $N_{\text{live}}{\le}K$
CrystalScore ($K{\le}9$)	**0.151** (F 1.00 / S 0.244 / St 0.619)	**0.938** (F 1.00 / S 0.938 / St 1.00)
Deployment status	**deployed**: real panels, live firewall, operating history	polygon-proven; real-market use fenced by the VoI gate

Table 1: The final comparison. The split verdict is part of the result: CRYSTAL-1 wins every command-surface axis *on the polygon*; R6c still leads as a deployed real-market system. The gap between them is exactly one port, and it is fenced by an honest gate, not by embarrassment (§7).

6 Composition: commands interact, so governance is machinery

Single commands behaving well does not imply command *programs* behave well. A 216-cell factorial over pairs of belief-dimension writes measured the interaction term directly: non-additivity is large (median $|\varepsilon| = 2.66$; 87% of cells material) and the cross-term *flips sign* across contexts (both + and − observed on two output channels; the effect survives raw pre-softmax logits, so it is not a renormalization artifact). Honest scope: the factorial explored corners *off* the visited joint manifold.

Consequently the write surface is governed by machinery, not etiquette: a per-principal **ledger** charges $D = \sum |\alpha|\tau$ plus a worst-case $\sum |\varepsilon|\tau$ (with $|\varepsilon|$ calibrated from the factorial — the default underestimated it $\sim$13×), a co-activation cap $N_{\text{live}} \le K$, a kNN **manifold governor** (flags 24/24 off-manifold corners at 1.2% held-out false-positive rate), and a **writ ladder** C0–C6 under which any version bump voids certification back to C1 — salami-slicing a big write into many small ones is caught by the ledger. The stack's own error rate has a first estimate: behavioral $\alpha_{\text{machine}} = 0.0$ (0/48; $\lesssim$ 6% by the rule of three, power 70.8%) — a proxy on the C1/C2 core, not yet the full-firewall quantity.

Joseph: this α_{machine} proxy is H9 in your program — the full-firewall estimate is open.

7 What did not work, and what bounds the claims

The certified tension. On the hard G12 substrate, raising legibility *costs* return and vice versa — certified as a null through the project's gate (return and legibility in strict tension). Anyone claiming interpretability is free on all tasks is measuring

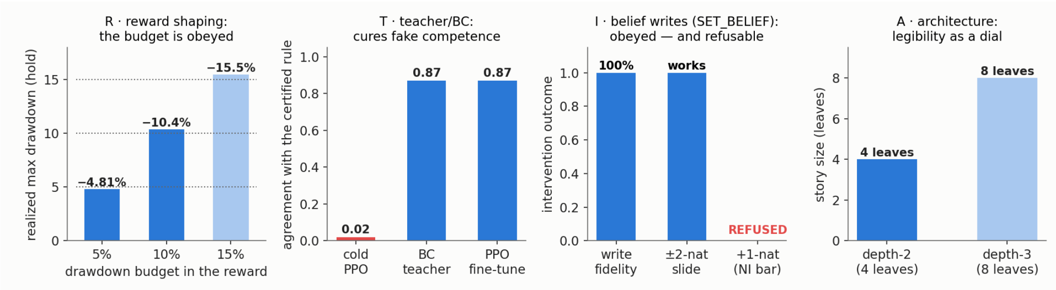

Figure 3: The four levers, one panel each. R: the risk-targeted reward $r = \text{pnl} - \lambda \max(0, \text{DD} - \text{budget})$ moves realized drawdown onto its budget (a 5% budget delivers −4.81% hold max drawdown; a 10% budget −10.4%; dotted lines = budgets). T: cold PPO produced an argmax over nearly flat preferences — competent-looking behavior from an effectively untrained net — cured by behavior cloning from a certified rule (87% agreement) that *persists* through PPO fine-tuning. I: `SET_BELIEF` writes land with fidelity 1.0 and a ±2-nat nudge slides behavior between named codes, while a +1-nat promotion was *refused* by the non-inferiority bar — the honest outcome. A: tree depth is a legibility dial (depth-2 = a 4-leaf story).

easy tasks.

The polygon→market gap. Every command-surface number in §3–§6 is polygon-proven (K=2, Series-G). On real daily markets the value of the belief information is honestly gated: the VoI gate opens on an engineered-positive substrate (VoI 5.92, +282% of blind) and *closes* on a real book (crypto VoI 0) — the regime is priced. Until a real VoI>0 substrate is found, CRYSTAL-1 on real markets is a transparency and monitoring object; its one real-panel deployment (a belief-driven exposure mode) roughly halves Dow drawdown but is enable-table-conditional. R6c keeps the deployed-system crown (Table 1); the split verdict *is* the honest result.

Post-hoc was not worthless. CHRL's primitive-discovery program produced certified frozen-test lifts (§2) and, more importantly, the concept inventory (motifs, codes, counterfactual replay) that CRYSTAL-1 promoted from probes into architecture. The right summary is not "post-hoc failed" but "post-hoc found the load-bearing concepts, and building them in made them 6× more compressible."

Assorted falsifications. The naïve entropy gate for command validation points backwards (trusts confident lies). A single-Gaussian manifold governor failed (multimodal cloud); kNN replaced it. The first α_{machine} estimate conflated effect-wrongness with optimality-wrongness and was corrected by adversarial review. Each is in the experiment logbook with its command and seed.

8 Related work

Interpretability-by-construction vs post-hoc explanation [Rudin, 2019, Lipton, 2018]; rigor and evaluation of interpretability [Doshi-Velez and Kim, 2017, Hoffman et al., 2018]; faithfulness as a criterion [Jacovi and Goldberg, 2020]. Belief-state control [Kaelbling et al., 1998] and regime models [Hamilton, 1989] supply the named memory; soft trees [Frosst and Hinton, 2017] the legible head; MDL [Rissanen, 1978]

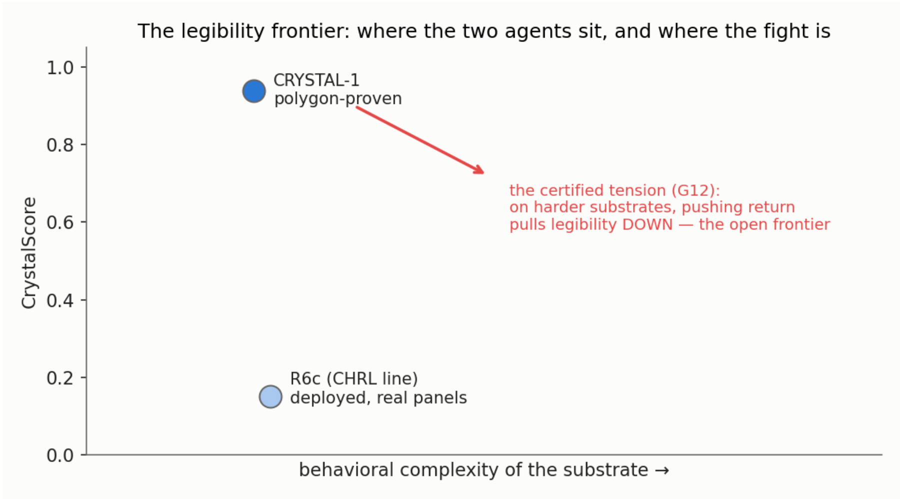

Figure 4: The honest frontier. CRYSTAL-1's ceiling metrics are real *and* measured on a substrate where legibility is cheap. The certified G12 result is that on harder substrates the return–legibility trade-off is a genuine tension, not an engineering artifact: that arrow is the interpretability track's open frontier, not a footnote.

the non-saturating ruler; interventional causality [Pearl, 2009] the faithfulness protocol. Hierarchical RL as partial legibility [Sutton et al., 1999, Bacon et al., 2017]; PPO and constrained variants [Schulman et al., 2017, Achiam et al., 2017]; goal-conditioned value functions [Schaul et al., 2015]. Influence machinery: reward shaping [Ng et al., 1999], imitation-then-optimize [Ross et al., 2011, Rusu et al., 2016]. The self-improvement layer that motivates all of this — LLM proposers over verified evaluators [Romera-Paredes et al., 2024, Novikov et al., 2025, Ma et al., 2024, Wang et al., 2023] and the statistics of adaptive search over noisy financial evaluators [Bailey and López de Prado, 2014] — is treated in the companion paper (*Hello Crystal*); this paper builds the object those loops are allowed to modify.

GAP: position vs concept-bottleneck models (Koh et al. 2020) and vs recent transformer interpretability (circuits) — one paragraph each, Joseph's lit pass.

9 The research program: twelve hypotheses, each with a kill test

This section is deliberately ambitious — the point of a legible, steerable, self-modifying agent is not one trading policy but a *general method for keeping evolving systems readable.* Each hypothesis is falsifiable; each names the observation that kills it.

Joseph: this program is yours to reshape: pick one hypothesis per month, or replace them with better ones — the kill-test discipline is the only non-negotiable.

1. **H1 — Machine simulatability predicts human simulatability.** The 0.94 tree agreement is machine-measured. *Test:* psychophysics — humans predict the policy's action in unseen cells. *Kill:* human accuracy at chance while machine simulatability stays at ceiling.
2. **H2 — A legibility scaling law.** Vocabulary size K tracks substrate behavioral complexity (the observed $C^* \approx K$ pattern) across substrates — a "Kolmogorov frontier" for policies. *Kill:* substrates where optimal K is unrelated to measured complexity.
3. **H3 — The return↔legibility tension is a substrate property, not an architecture property.** Prediction: the tension vanishes on VoI>0 substrates

and is maximal where the regime is fully priced. *Kill:* tension persists on an engineered VoI>0 polygon.

4. **H4 — Interpretability drift.** Under the self-improvement loop, naming faithfulness decays unless actively re-certified ("semantic drift" of the belief coordinates). *Test:* track faithfulness across accepted loop generations. *Kill:* faithfulness stays at 1.0 over $\geq$20 accepted changes with no re-certification.

5. **H5 — Motif conservation across architectures.** CHRL's discovered primitives re-emerge as CRYSTAL-1 tree leaves (homologous behaviors across "species"). *Test:* match Stage-7 primitives to leaf semantics. *Kill:* no better-than-chance correspondence.

6. **H6 — The write protocol is pharmacological.** `SET_BELIEF` dose-response curves are monotone and saturating everywhere (polygon threshold 0.26 generalizes). *Kill:* non-monotone dose regions on any certified substrate.

7. **H7 — Sign-epistasis survives the deployed manifold.** The C-5 factorial explored off-manifold corners. *Test:* re-measure within the visited manifold (L1 $\leq$ 0.30). *Kill (of the governance premise):* on-manifold interactions are additive — then the ledger's ε term is precautionary overhead.

8. **H8 — The lie-detector generalizes.** Authority-tracks-NLL discriminates wrong-but-sharp commands on real panels at operationally useful rates. *Kill:* detection at chance off-polygon.

9. **H9 — $\alpha_{\mathbf{machine}}$, full-firewall.** The 0/48 C1/C2-core proxy holds ($\leq$ 5%) when estimated through the complete gate stack. *Kill:* full-stack estimate an order worse.

10. **H10 — Faithful client-facing explanations.** Every "why" sentence shown to a user can be made *measurably* faithful (the explanation gate): perturb the stated reason, behavior must move accordingly. *Kill:* explanations that pass fluency review but fail perturbation faithfulness at scale.

11. **H11 — Certified vocabulary growth.** `GROW_K` can be run as a certified architectural operation: the agent learns a new regime word without semantic drift of the old ones. *Kill:* any old-coordinate faithfulness drop after growth.

12. **H12 — The score applies to the improver.** CrystalScore's axes (faithfulness / simulatability / stability) transfer to the *coding agent's* own proposals — interpretability of the improver, not just the improved. *Test:* score proposal rationales against their measured effects. *Kill:* rationale-effect agreement at chance (the LLM's stated reasons are confabulation — itself a publishable result).

10 Limitations

(i) All command-surface results are K=2, Series-G polygon; the ceiling metrics are measured where legibility is cheap. (ii) Sign-epistasis is demonstrated off the joint manifold (H7 is the repair). (iii) $\alpha_{\mathrm{machine}}$ is a core proxy (H9). (iv) The $\varepsilon \to D$ units bridge in the ledger is a bookkeeping convention, not a validated equivalence. (v) The VoI-gate discrimination is n=1 engineered-positive vs n=1 real-negative. (vi) Human simulatability is untested (H1) — the metric that names the property is still machine-only.

Ivan: decide: does the B3 real-panel exposure mode belong here as a seventh limitation (enable-table conditionality) or in §7 only?

Acknowledgments and drafting discipline

Prepared *write-paper-first* [Eisner, 2010]: every margin `todo` marks a real gap. Experiments were run and logged under the repository's honesty contract — every entry names its null, its seed, and its worst caveat — in the spirit of Feynman [1974]: *"The first principle is that you must not fool yourself — and you are the easiest person to fool."* LLM coding agents (Claude, Codex) operated the loops under the stated controls; the agents propose, the gates dispose.

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